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# CAT 6 FLASHER 12 V REVERSED VOLTAGE TEST REPORT



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|---|---|---|---|
| **Part Name** | CAT 6 FLASHER 12 V | **Application** | COMMERCIAL APPLICATION |
| **Customer Part No.** | B00 861 544 00 32 | **Customer** | FORCE MOTOR |
| **UCAL Part No.** | 400230122002 | **Report No** | R&D/CAT6/12VF/DUR/26/037 |

## INTRODUCTION:

UCAL is currently supplying 12V CAT 6 Flashers to Force Motor for commercial applications. This validation plan has been prepared to define the test requirements and validation activities for the current production parts of the respective product. The testing will be carried out as per the approved validation plan to verify product performance, functional reliability, and compliance with specified requirements under defined test conditions.

## PURPOSE:

To evaluate the 12 V CAT 6 flashers assembly after **Reversed voltage test.**

## TEST SPECIFICATION:

Test standard as per **ISO 16750-2:2023**. DUT should operate within specified functional parameters.

## ACCEPTANCE CRITERIA:

The Electronic Component should function normally after the test (**Function Status A**). There should not be any deformation, discoloration, fire, smoke, or crack during and after the test..

## RESULT:

| SAMPLE NO | PERFORMANCE | APPEARANCE | JUDGEMENT |
|---|---|---|---|
| S:03 | O | O | O |
| S:04 | O | O | O |

**O: OK** **X: NG** **∆: To be confirmed**

## CONCLUSION:

The performance of the Flasher assembly is acceptable after test. The samples are met the requirements.

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|---|---|
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| **Prepared By** | **Approved By** |

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DATE:15.05.2026



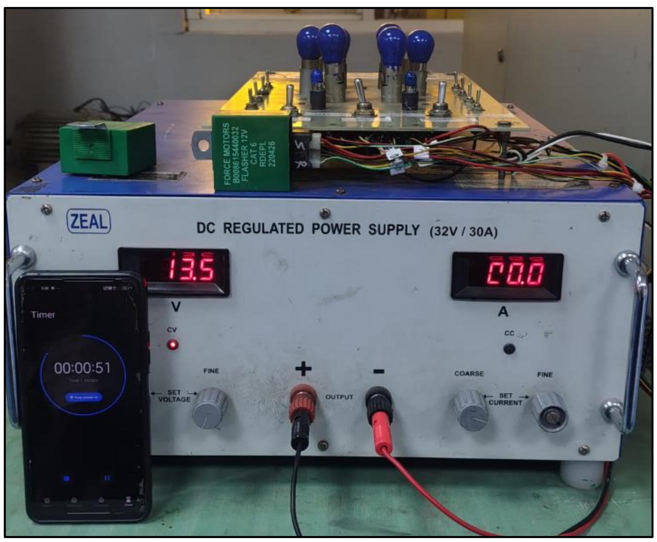
## PERFROMACE TEST SPECIFICATIONS:

| SNO | PARAMETERS | SPECIFICATIONS |
|---|---|---|
| 1 | Operating  voltage | 10~16  V |
| 2 | Test voltage | 13.5V |
| 3 | Flash rate(Turn signal/Hazard warning | 60~120 Flashes/Minute |
| 4 | One lamp failure flashing rate | >140 Flashes/Min(or)1.8 times of flash rate whichever is greater |
| 5 | Duty Cycle | 40% ~60% |

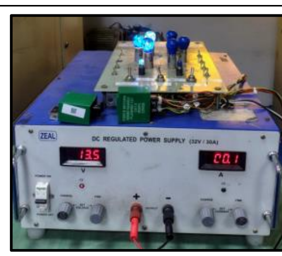
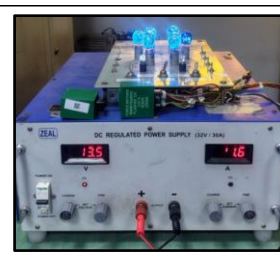
TEST PROCEDURE:

Connect the Electronic Component to the power supply, input and loading devices. The power supply device should be reversely connected to the Electronic Component. Apply supply voltage **VA (-13.5±0.5V)** for **1 minute**.

## TEST SETUP




## TURN SIGNALS & HAZARD SIGNALS:

| LEFT SIGNAL | RIGHT SIGNAL | HAZARD SIGNAL |
|---|---|---|
|  |  |  |

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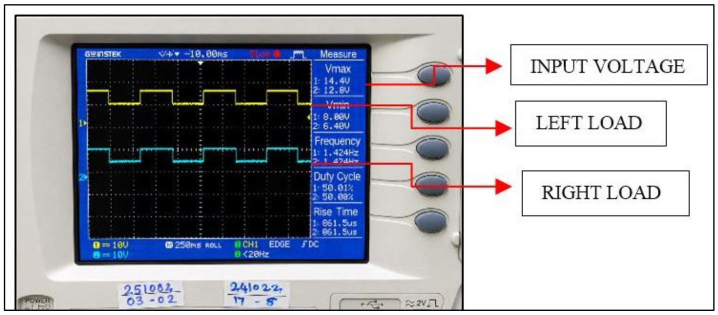
## WAVEFORM DATA:




## TEST OBSERVATION:

| Description | Load<br>LED& CLUSTER | Observations(at 12 V)<br>Sample No:03&04 | Remark |
|---|---|---|---|
| Turn Signal RH | 3 x 21 W Lamp + 1x 2W LED | 85 | OK |
| Turn Signal LH | 3 x 21 W Lamp + 1x 2W LED | 85 | OK |
| Hazard Mode | 6 x 21 W Lamp + 2x 2 W LED+ | 85 | OK |
| Fault condition Any one 21W Lamp failure | 2 x 21 W Lamp + 1x 2 W LED | 170 | OK |
| Duty cycles for RH&LH | 3 x 21 W Lamp + 1x 2 W LED | 50% | OK |
| Duty cycles for Hazard Mode | 6 x 21 W Lamp + 1x 2 W LED | 51% | OK |

NOTE: The observed flashes are within the specifications & the samples met the performance requirements.

## TESTED SAMPLES PHOTOS:

| BEFORE TEST SAMPLE – S:03,04 | AFTER TEST SAMPLE – S:03,04 |
|---|---|
|  |  |

## OBSERVATION:

There is no significant change in the electrical performance of the flasher samples. The functionality remains satisfactory. The observed flashes are within the specified requirements & the samples met the performance requirements. All function of the Flasher performs as designed before & after the test.